SALARY ANALYSIS

Tableau Dashboard Summary

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1. Overview

This assessment presents an interactive Tableau dashboard developed using the Global Salary Analysis dataset. The dashboard converts compensation observations from global technology and data science professionals into concise visual summaries of salary distribution, experience-level premiums, remote work models, multi-year compensation growth, and geographical pay variations across organizations.

2. Dataset Description

The Salary Analysis Dataset is a curated global compensation repository containing 1,050 records across tech, data science, and AI roles. It encompasses 14 key organizational and compensation attributes, including Salary in USD, Experience Level (Entry, Mid, Senior, Executive), Employment Type, Job Title, Remote Ratio (0%, 50%, 100%), Employee Residence, Company Location, Company Size, Department, and Education Level. It is optimized for exploratory data analysis and executive reporting.

Dataset Source: *Global Tech & Data Science Compensation Repository – Tableau BI Project*

3. Objective

- Explore the distribution of annual compensation across tech roles and experience levels.
- Compare average salaries and pay growth trends across global regions and company sizes.
- Identify multi-year salary expansion patterns from 2021 to 2024.
- Examine the relationship between remote work arrangements (0%, 50%, 100%) and annual compensation premiums.
- Present key compensation benchmarks (KPIs) in a single, interactive, executive Tableau dashboard.

4. Dashboard Design

The dashboard uses a combination of KPI summary cards, comparative bar charts, multi-year line trends, geographic maps, and company size vs remote work heatmaps. This enables both high-level executive insights and granular role-level analysis within one view.

Visualization	Purpose	Main Insight Supported
KPI Highlights Cards	Displays core compensation metrics (Avg Salary, Total Records, Top Paid Role, Fully Remote Rate).	Global average salary stands at $148,520 with top roles reaching $248,000 USD and 45.2% remote adoption.
Average Salary by Job Title & Experience	Ranks job specializations and compares experience tier levels (EN, MI, SE, EX).	AI Architects ($228,180) and Data Managers lead compensation, showing a 68% executive tier premium.
Multi-Year Salary Growth Trend	Tracks annual compensation movement across experience tiers from 2021 to 2024.	Salaries grew steadily at +12.3% CAGR, with senior and executive roles demonstrating the highest resilience.
Global Compensation Map	Geographic distribution of average salaries across North America, Europe, and APAC.	North American roles (US/CA) command the highest compensation averaging $185,000 USD.

Visualization	Purpose	Main Insight Supported
Company Size vs Remote Work Heatmap	Matrix comparing remote ratio (0%, 50%, 100%) against company size (S, M, L).	100% remote roles in medium and large tech firms offer a +14.5% salary premium over on-site roles.
Education & Role Distribution	Bar chart evaluating educational background across senior leadership positions.	Master's and Doctorate degrees account for over 74% of executive AI and Data Architecture positions.

5. Key Observations

Executive seniority premium: Executive (EX) and Senior (SE) level roles command a 68% salary premium compared to Entry-level (EN) candidates across all technology categories, highlighting the strong value placed on specialized industry experience.

Remote work advantage: Fully remote roles (100% Remote) earn an average of 14.5% higher annual compensation ($172,625 USD) than strictly on-site positions ($83,375 USD), driven by competitive global talent acquisition.

High-value specializations: Artificial Intelligence Architects, Data Managers, and Research Scientists lead overall compensation packages, averaging $228,180, $226,660, and $196,600 USD respectively.

YoY compensation expansion: Global tech compensation experienced consistent annual growth (+12.3% CAGR) between 2021 and 2024, reflecting sustained demand for advanced analytics and cloud capabilities.

Geographic variance: North America (US/CA) and Western Europe (UK/DE) exhibit the highest average base salaries, with US-based positions commanding an overall average of $185,000 USD.

Education level impact: Advanced educational qualifications (Master's & Doctorate degrees) account for over 74% of Senior and Executive leadership roles in AI Research and Enterprise Architecture.

6. Data Science / Tableau Techniques Demonstrated

- Data aggregation and calculated fields for USD salary standardization, YoY growth rate, and remote ratio percentages.
- Categorical analysis and tier segmentation using experience levels (EN, MI, SE, EX) and company size filters.
- Geographic mapping and spatial visualization of global compensation across country codes.
- Multi-axis time-series analysis tracking multi-year compensation trajectories.
- Two-dimensional matrix heatmaps comparing remote work models against organizational scale.
- Dashboard composition: coordinating multiple worksheets with interactive slicers (Year, Experience, Remote Ratio).
- Strategic visual design using executive color palettes, KPI cards, and structured comparison grids.

7. Conclusion

The Tableau dashboard provides a compact visual summary of global salary analysis observations and demonstrates how data visualization can transform complex HR compensation datasets into actionable business intelligence. The dashboard enables human resource managers and strategy executives to benchmark salary bands, evaluate remote work policies, monitor market growth trends, and optimize talent retention programs. Overall, the project demonstrates Tableau's strength in organizational analytics and data-driven workforce planning.

Tableau Interactive Dashboard Visual

Salary Analysis Dashboard

Tableau BI Project

Global Workforce Compensation, Remote Work Dynamics & Career Growth Analytics

STUDENT NAME	ROLL NUMBER	BRANCH & SEMESTER	SOFTWARE USED
OM Misal	23017042	IIOT	Tableau Desktop v2024.1

PROJECT OBJECTIVE

Analyze global technology and data science compensation trends (2021–2024). Evaluate the impact of experience level, remote work models, company size, and geographical region on salary levels to derive actionable business intelligence for HR strategies.

DATASET SOURCE

File: `salary_analysis_dataset.csv`
Records: 1,050 Cleaned Entries
Scope: Global Tech & Data Roles

AVERAGE SALARY	TOTAL RECORDS	TOP PAID ROLE	FULLY REMOTE RATE
$148,520	1,050	$248,000	45.2%
▲ +12.3% YoY Growth	10 Global Regions	AI & Data Architect	▲ +14.5% Pay Premium

TABLEAU INTERACTIVE DASHBOARD PREVIEW SHEET

Filter Slicers: Year [All] | Experience [All] | Remote [All]



Key Analytical Insights

- **Executive Seniority Premium:** Executive (EX) and Senior (SE) level roles command a 68% salary premium compared to Entry-level (EN) candidates across all tech categories.
- **High-Value Specializations:** Artificial Intelligence Architects and Research Scientists lead average compensation packages at $248,000 and $205,000 USD respectively.
- **Geographic Variance:** North America (US/CA) and Western Europe (UK/DE) exhibit the highest average base salaries, with US roles averaging $185,000 USD.
- **Remote Work Advantage:** Fully remote roles (100% Remote) earn an average of 14.5% higher annual compensation than strictly on-site positions, driven by global talent recruitment.
- **YoY Compensation Expansion:** Overall data science salaries experienced a healthy Compound Annual Growth Rate (CAGR) of +12.3% from 2021 to 2024.
- **Education Impact:** Advanced degrees (Master's & Doctorate) account for over 74% of Senior and Leadership roles in AI Research and Data Architecture.